

Introduction

1 MRI reconstruction and its role in clinical practice

The goal of Magnetic Resonance Imaging (MRI) reconstruction is to convert the measured raw MR data into images that can be clinically interpreted, and thus it plays a fundamental role in the clinical use of MRI.

In the early days of MRI, most of the research focused on hardware improvements and the development of new pulse sequences aiming at faster acquisition and/or improved contrast, while the image reconstruction was performed via a simple Fourier transform. Interestingly, the Fourier transform was not the first approach to MRI reconstruction. In 1973, Paul Lauterbur published his seminal paper showing the first MR image of two water-filled glass tubes. This image was obtained by applying magnetic field gradients at several different directions and performing a reconstruction similar to the filtered back projection used in computed tomography. Although this was very exciting, the reconstruction was very time consuming and the images were prone to artifacts, mainly due to the main magnetic field inhomogeneities. The introduction of spin warp imaging, in which data are acquired on a uniform grid in k-space (Cartesian sampling), enabled the application of the fast Fourier transform (FFT) for MR image reconstruction. This made MR reconstruction faster and less sensitive to field inhomogeneities, which were crucial for the clinical adoption of MRI.

The fast Fourier transform has well served the MR community as the conventional image reconstruction method for fully sampled Cartesian k-space. However, it is important to mention that it is an approximation that does not always describes the MR physics very well, which may lead to artifacts in the reconstructed images. Moreover, it requires the acquisition of a fully sampled Cartesian k-space, which may lead to very long scan times. In other instances, such as when only undersampled k-space data are available, when data are acquired with multiple coils or on a non-Cartesian grid, or when physical effects like field inhomogeneities are important, the simple FFT image reconstruction becomes suboptimal. In these scenarios, advanced reconstruction methods that use more accurate physics models for the data acquisition or apply prior knowledge about the image properties are more appropriate.

An important conceptual shift that facilitated the development of such advanced MR reconstruction methods has been the move towards model-based reconstruction that uses an inverse problem formulation to provide a unified view of different reconstruction problems. This also enables the application of algorithms for solving these inverse problems with proven convergence guarantees. It also allows the use of more complicated MR physics models to either reduce artifacts or extract physical properties of the tissue from the measurement data (quantitative MRI). In this optimization framework, it is easy to include additional constraints, e.g., priors describing image properties or structure. This framework is very powerful for the development of new reconstruction approaches and understanding the limitations arising from various approximations.

Although advanced MR reconstruction techniques have a long history, the last two decades marked a fast development of this research field, as well as the adoption of advanced reconstruction methods by vendors and subsequent integration into clinical practice. Parallel imaging has revolutionized MRI

by providing typically a two–three-fold acceleration, reducing the risk of motion artifacts, improving patient comfort, and enabling new applications. With the introduction of compressed sensing, an additional acceleration was achieved with virtually no loss in image quality which has already helped many hospitals to improve their productivity by scanning more patients a day. More recently, deep learning-based reconstruction approaches, using data adapted priors that can push the acceleration even a little further, have gained a lot of attention. These advances were due to a combination of factors, including the advances in computation speed, development of the necessary mathematical frameworks, and advances in deep learning, as well as the clinical need for improved imaging speed.

2 Organization of the book

This book covers the fundamental concepts of MR image reconstruction, including its formulation as an inverse problem and the most common models and optimization methods used to reconstruct MR images nowadays. The most recent developments, such as compressed sensing, tensor-based reconstruction, and machine learning-based reconstruction are also included. Approaches for specific applications such as non-Cartesian imaging, undersampled reconstruction, motion correction, dynamic imaging, and quantitative MRI are discussed. Code examples and tutorials are provided for some of the MRI reconstruction methods discussed in this book. The tutorials can be accessed at <https://www.elsevier.com/books-and-journals/book-companion/9780128227268>.

Chapters 1–4 describe the basic principles of MR image reconstruction and are complemented by Appendix A, in which the necessary mathematical tools are explained in more detail. Chapter 1 gives a brief introduction to MR physics and the image formation process. Chapter 2 formulates MRI reconstruction as an inverse problem and discusses some basic considerations for solving this inverse problem. Chapter 3 gives an overview of some commonly used algorithms used for MRI reconstruction. Chapter 4 considers sampling on a non-Cartesian grid and the corresponding reconstruction methods.

Chapters 5–11 describe more advanced reconstruction methods to speed up MR acquisition. Chapter 5 provides an overview of early constrained image reconstruction methods used in MRI. Chapter 6 explains the basics of parallel imaging reconstruction and discusses several approaches for volumetric, dynamic, and non-Cartesian acquisitions. In Chapter 7 simultaneous multislice (SMS) MRI reconstruction that enables acquisition of multiple slices at the same time is discussed. Chapter 8 presents an overview of compressed sensing and its application in rapid MRI by exploiting image sparsity. Low-rank matrix and tensor approaches to MR image reconstruction are discussed in Chapter 9. In Chapter 10 undersampled MRI reconstruction methods that approximate groups of signals by subspaces, dictionary learning methods, and smooth manifold based approaches are described. Chapter 11 provides the basics of machine learning for MRI reconstruction and discusses the use of various neural networks for single and multicoil MRI reconstruction.

Model-based MRI reconstruction approaches are discussed in Chapters 12–16. Chapter 12 explains the imaging impacts of magnetic field inhomogeneities and describes approaches that incorporate models of field inhomogeneity in the image reconstruction to correct for it. Similarly, since motion can introduce artifacts in the reconstructed images, Chapter 13 describes how to incorporate motion into the reconstruction process to correct for it. In Chapter 14, chemical-shift encoding methods, together with reconstruction approaches to separate water and fat signals, are described. Chapter 15 discusses various reconstruction approaches for model-based quantitative parameter mapping, while Chapter 16 describes specific approaches for quantitative susceptibility mapping reconstruction.